

Financial Infrastructure Research Series

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Working Paper

FOUNDATIONAL FRAMEWORK · JULY 2026

Semantic Tokenization for Financial Infrastructure: A Foundational Framework

*Canonical Identity, Multi-Rail Settlement, and Representation Layers
for the Next Generation of Financial Market Infrastructure*

Classification: Public Research Working Paper

Audience: Financial Infrastructure Architects · Central Bank Technologists · FMI
Operators · Tokenization Platform Engineers · Policy Researchers

Date of Issue: July 15, 2026 | Version 1.0

Keywords: Semantic Tokenization · Canonical Identity · Multi-Rail Settlement · Financial Market
Infrastructure · Representation Layer · Compliance Semantics · Cross-Rail Interoperability · CBDC
· ISO 20022 · DLT Post-Trade

Abstract

The rapid proliferation of distributed ledger technology (DLT)-based tokenization platforms, central bank digital currency (CBDC) rails, and legacy central securities depository (CSD) systems has generated a fundamental architectural crisis in financial market infrastructure: the same underlying asset is simultaneously represented as syntactically divergent, semantically uncoordinated token objects across multiple incompatible systems. This fragmentation produces settlement failures, compliance duplication, cross-rail identity ambiguity, and the systematic erosion of audit trail integrity — outcomes antithetical to the resilience and efficiency objectives that tokenization was intended to advance.

This paper introduces **Semantic Tokenization** as a foundational architectural discipline for the next generation of financial market infrastructure. Semantic Tokenization reconceptualizes the financial token not as a technical ledger artifact but as a *meaning-bearing, identity-anchored, compliance-aware, settlement-deterministic unit of financial representation*. The framework comprises five interlocking components: (1) the **Canonical Identity Framework**, providing rail-agnostic persistent asset identifiers; (2) the **Unified Representation Model (URM)**, enabling meaning-preserving translation across heterogeneous data paradigms; (3) the **Multi-Rail Deterministic Settlement model**, embedding settlement semantics intrinsically within the token rather than in operator configuration; (4) the **Semantic Compliance Routing Engine (SCRE)**, encoding jurisdiction-aware compliance predicates as first-class token properties; and (5) the **Cross-Rail Semantic Interoperability Protocol**, guaranteeing that asset identity, compliance state, and settlement semantics are fully preserved across rail transitions.

The paper further presents a complete reference architecture for a Semantic Tokenization Platform (STP), discusses governance and standards alignment pathways with ISO 20022, GLEIF LEI, and proposed Canonical Asset Identifier (CAI) standards, and outlines a phased adoption roadmap for FMIs, regulators, and tokenization platforms. The implications for CSDs, CCPs, CBDC platforms, and regulatory oversight frameworks are examined in detail. The central conclusion is that tokenization without semantic coherence produces syntactic fragmentation

rather than infrastructure modernization, and that the industry must establish canonical identity and compliance semantic standards before tokenized markets reach systemic scale.

1. Introduction and Motivation

The past decade has witnessed an unprecedented proliferation of financial tokenization initiatives. Central banks across more than fifty jurisdictions are actively piloting or deploying CBDC platforms. Major central securities depositories — from Euroclear and Clearstream to the Depository Trust Company (DTC) — are conducting DLT integration experiments. Private tokenization platforms have tokenized government bonds, equities, real estate investment trust shares, trade receivables, and money market fund units across a dozen competing blockchain protocols. The Financial Stability Board estimated in 2024 that while tokenization remains small in aggregate scale, its growth trajectory is accelerating, with tokenized money market products alone attracting significant institutional inflows from 2023 onward.

Yet beneath this surface of innovation lies a structural incoherence that the industry has not adequately confronted: the financial token, as currently conceived and implemented, is a *syntactic* artifact. It carries transfer logic, balance data, and perhaps rudimentary access control. It does not carry — in any standardized, interoperable, machine-readable form — the full legal meaning, compliance state, settlement semantics, or identity anchoring that its underlying real-world asset possesses. The consequence is that the same government bond can simultaneously exist as an ERC-1400 token on a permissioned Ethereum network, a UTXO-structured record in a CBDC settlement system, a CSD account-based position in a legacy T2S environment, and a payment obligation in an ISO 20022 message queue

— with no authoritative mechanism to assert that these four representations are the same instrument, or to propagate lifecycle events, compliance state changes, or settlement obligations coherently across all four.

This is not merely a technical interoperability problem. It is a *semantic architecture deficit*. The current generation of tokenization projects has conflated the digitization of assets with the semantic transformation of assets. Digitization moves a record from one medium to another. Semantic transformation encodes the full meaning of an asset — its legal identity, its compliance obligations, its settlement determinism, its representational integrity across systems — into the digital object itself.

This paper makes the following central argument: **Semantic Tokenization** — the discipline of encoding the full legal, economic, compliance, and settlement meaning of a financial asset into its digital representation as intrinsic properties — is the missing architectural layer that must be established before tokenized financial markets can achieve the systemic reliability, regulatory integrity, and cross-platform interoperability required for institutional adoption at scale.

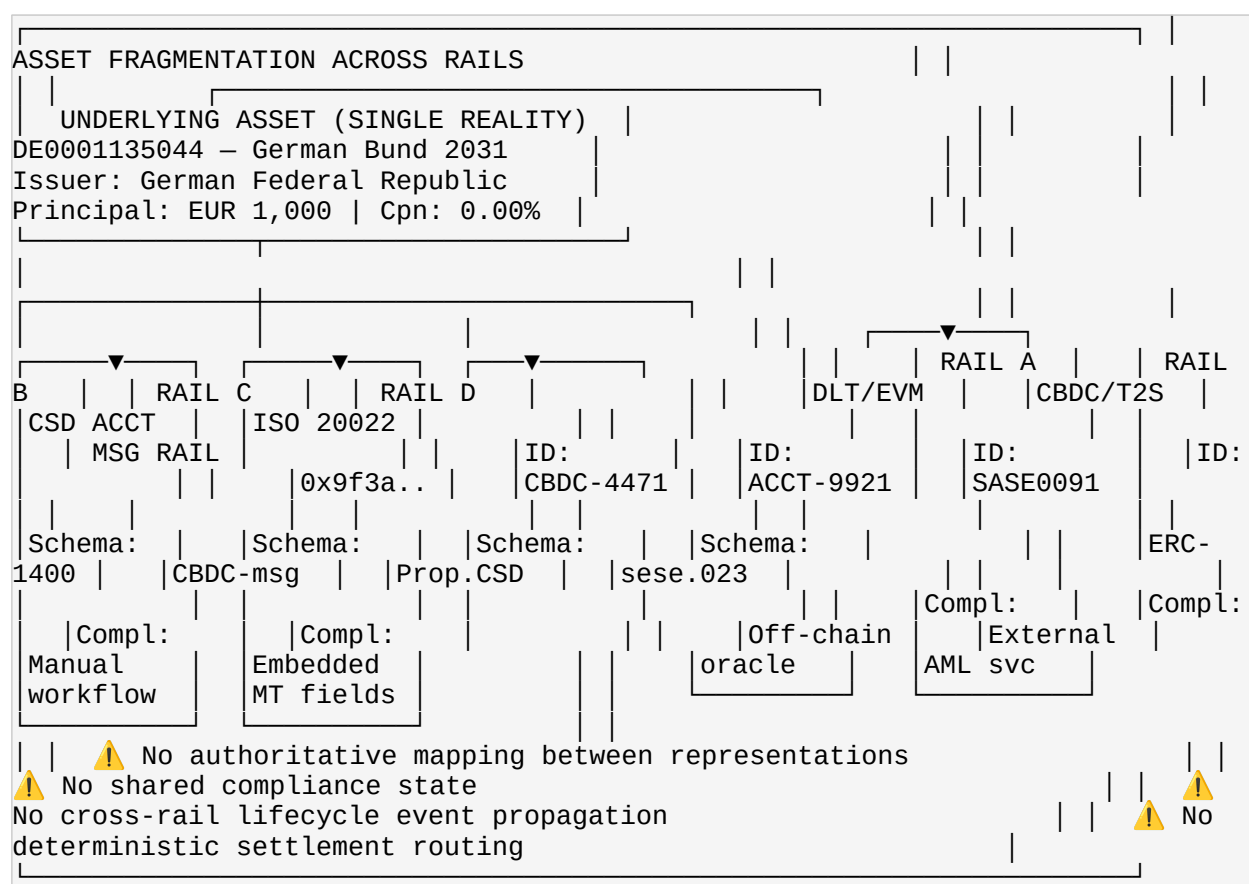
The paper proceeds as follows. Section 2 diagnoses the four principal failure modes produced by the current syntactic tokenization paradigm. Section 3 introduces Semantic Tokenization and its five-layer architecture. Sections 4 through 8 develop each architectural layer in technical depth: Canonical Identity, Unified Representation, Multi-Rail Settlement, Semantic Compliance Routing, and Cross-Rail Interoperability. Section 9 examines implications for the principal classes of financial market infrastructure. Section 10 presents a complete reference architecture. Section 11 addresses governance, standards alignment, and adoption pathways. Section 12 concludes.

2. Problem Statement

2.1 The Fragmentation Problem

A sovereign government bond is a legally defined instrument: it has a single issuer (a national treasury), a single principal legal jurisdiction, a globally recognized identifier (an ISIN assigned by a national numbering agency), a fixed set of cash flow obligations, a defined maturity, and a regulatory classification under the applicable securities law of its issuing jurisdiction. These properties are invariant across representations — they attach to the underlying instrument, not to any particular record of it.

In the current tokenization landscape, this single instrument may generate four or more independent digital representations, each encoded in a different schema, carrying a different identifier, managed by a different custodian or platform, and subject to different compliance checks. The diagram below illustrates this fragmentation schematically:



The BIS CPMI has observed that the lack of interoperability among DLT platforms and between DLT platforms and traditional financial infrastructure constitutes one

of the principal barriers to meaningful tokenization scaling. This fragmentation is not merely an inconvenience — it creates systemic risk exposure as the volume of cross-rail asset movements grows.

2.2 Identity Collapse Under Multi-Rail Conditions

Without a canonical identity layer, a bond token on Rail A and the same bond token on Rail B are, from the perspective of any system that receives them, entirely unrelated objects. They share no authoritative common identifier, no shared state, and no binding protocol that asserts their equivalence. This "identity collapse" under multi-rail conditions produces four categories of operational failure:

- **Double-settlement risk:** Settlement systems on different rails may independently initiate settlement of the same obligation, having no mechanism to detect that the counterparty position on Rail B is the same instrument already settled on Rail A.
- **Compliance duplication and gaps:** KYC, AML, and sanctions screening may be performed independently on each rail without shared state, producing both redundant checks and dangerous gaps when the compliance clearance on one rail is not propagated to another.
- **Audit trail breaks:** Regulatory examination of a transaction's full lifecycle is impossible if each rail maintains an independent, unlinked record. The transaction's economic substance — a transfer of a specific instrument from one counterparty to another — cannot be reconstituted from fragmented rail-specific logs.
- **Liquidity fragmentation:** Without identity resolution, netting systems cannot recognize that positions across rails represent the same underlying instrument, preventing multilateral netting that would otherwise reduce gross settlement obligations.

2.3 Compliance Semantics Gap

The dominant token standards in use today — ERC-20 and ERC-1400 for Ethereum-compatible networks, bespoke CBDC payload schemas, and ISO 20022 message structures for traditional rails — treat compliance as an *extrinsic* property.

Compliance checks are performed at the application layer by external services (oracle contracts, off-chain AML engines, manual workflow systems) that interrogate the token and its proposed transaction and return a boolean authorization. The compliance logic itself is not embedded in the token's semantic definition; it is attached procedurally by the platform operator.

This architecture has three structural deficiencies. First, compliance logic is platform-specific and non-portable: a token that clears compliance on Platform A carries no attestation that is recognized by Platform B. Second, compliance state is not versioned or anchored to the token: if a counterparty is added to a sanctions list after a token was originally compliance-cleared, there is no systematic mechanism to propagate that state change to all representations of the asset across all rails. Third, the compliance schema is not machine-readable by regulators: supervisors must reconstruct compliance status from transaction logs rather than querying a structured, semantically rich compliance record embedded in the asset itself.

2.4 Settlement Determinism Failure

In the current paradigm, settlement routing is determined by operator configuration: a custodian or platform operator decides, based on business rules and bilateral agreements, which settlement rail to use for a given transaction. This produces what we term *settlement non-determinism*: given the same asset and the same transaction intent, different operators may route settlement differently, producing inconsistent outcomes including failed Delivery versus Payment (DVP) across rail boundaries, timing ambiguity between delivery and payment legs, and non-atomic settlement that leaves counterparties exposed to principal risk during the settlement window.

Core Diagnostic

The fundamental problem is not that tokenization platforms are poorly engineered. It is that the token, as a data structure, is semantically impoverished. It carries what is necessary to transact on a single platform, but not what is necessary to exist coherently across an ecosystem of platforms. Semantic Tokenization addresses this deficit at the architectural level.

3. Semantic Tokenization: Core Concepts

Primary Definition

"Semantic Tokenization is the discipline of encoding the full legal, economic, compliance, and settlement meaning of a financial asset into its digital token representation — such that the token carries self-describing identity, jurisdiction-aware compliance semantics, multi-rail settlement instructions, and a canonical cross-system identifier as intrinsic attributes, making the token's meaning independently verifiable, computationally interpretable, and preserved across rail transitions without loss of semantic fidelity."

3.1 From Syntactic to Semantic Tokens

The distinction between a syntactic token and a semantic token is architectural, not incremental. A syntactic token is a data object whose schema encodes the minimum state required for valid transactions on a specific platform: an owner address, a balance, and transfer logic. Its meaning — what asset it represents,

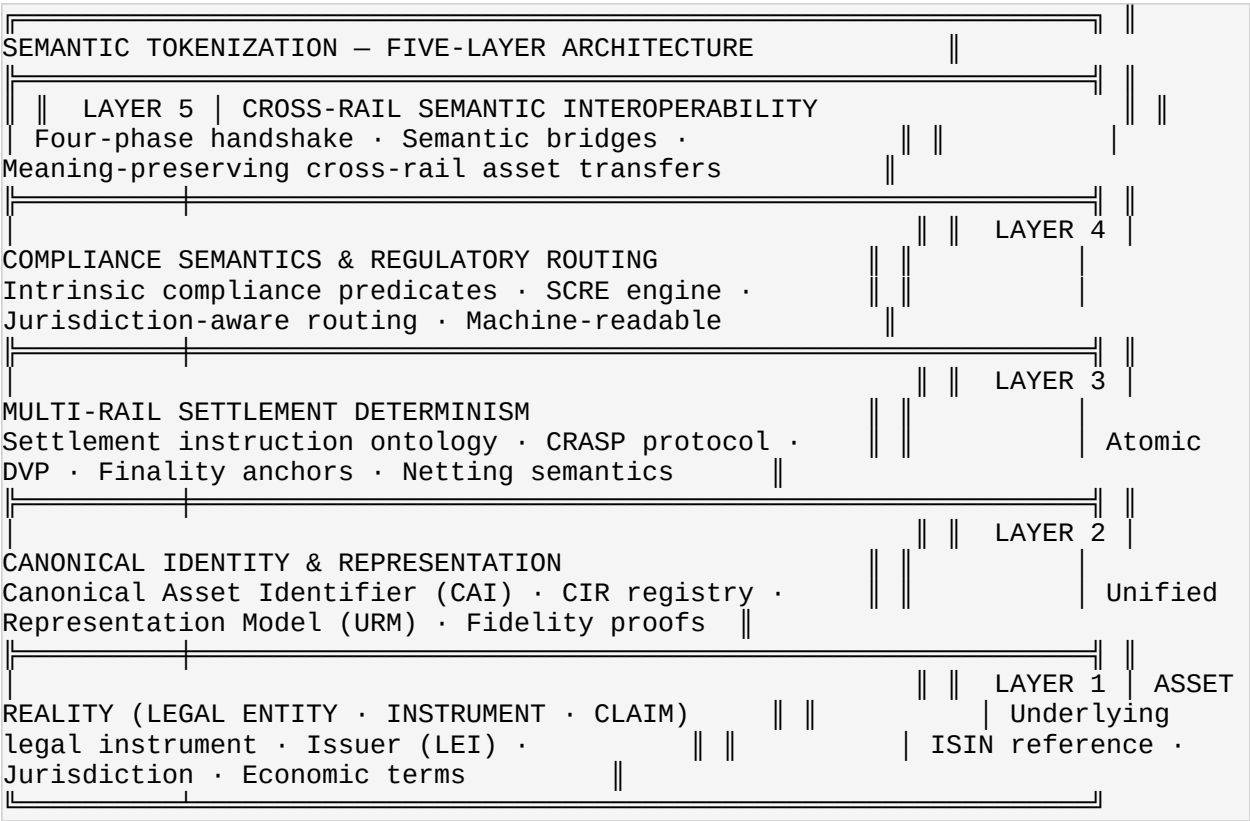
under what legal conditions, with what settlement obligations — is not encoded in the token itself but is assumed by the platform that hosts it.

A semantic token, by contrast, encodes its own meaning as a first-class attribute of its data structure. It is self-describing: any system that receives a semantic token can, without consulting external platform-specific configuration, determine what real-world asset it represents, what compliance checks govern its transfer, how it must settle, and what jurisdictions' laws apply to it. The table below formalizes this distinction across five dimensions:

Dimension	Syntactic Token	Semantic Token
Identity	Platform-local address or token ID; no cross-rail resolution	Canonical Asset Identifier (CAI) bound to a global Canonical Identity Record; rail-agnostic resolution
Compliance	Extrinsic; enforced by platform-level access control or external oracle; not portable	Intrinsic; jurisdiction-aware compliance predicates embedded as structured schema; machine-readable by regulators
Settlement	Operator-configured; determined at transaction time by platform business rules	Semantically embedded; settlement rail, DVP requirements, finality anchor, and netting eligibility are token properties
Interoperability	Syntactic bridges move token data between rails; semantic meaning may be lost or distorted in translation	Semantic bridges translate meaning with fidelity guarantees; canonical identity and compliance state are preserved and attested
Lifecycle Management	Platform-specific lifecycle events (maturity, recall, suspension) are not automatically propagated to other rail representations	Lifecycle events are registered against the Canonical Identity Record and propagated to all rail representations via the Cross-Rail Semantic Protocol

3.2 The Five Semantic Layers

Semantic Tokenization is structured as a five-layer architecture, each layer providing a distinct and necessary semantic capability. The layers are interdependent: each higher layer depends on the services provided by the layers below it. The architecture is illustrated below:



Layer 1 is the ground truth: the real-world asset with its legal, economic, and jurisdictional properties. Layer 2 creates and maintains the canonical digital identity that binds all representations to this ground truth, and provides the Unified Representation Model for cross-paradigm translation. Layer 3 embeds settlement instructions within the token as semantic properties. Layer 4 encodes compliance obligations intrinsically. Layer 5 coordinates cross-rail asset movements with full semantic preservation. Each layer is developed in full in Sections 4 through 8 below.

4. Canonical Identity Framework

4.1 The Canonical Identity Problem

Definition

"Canonical identity" is a globally unique, rail-agnostic, persistent identifier that resolves to the same underlying real-world asset regardless of which rail, platform, or jurisdiction is querying it, and that is bound to a structured semantic record encoding the asset's full legal, compliance, and settlement properties in their authoritative form.

Existing identifier systems provide partial solutions but are structurally inadequate for semantic tokenization. The International Securities Identification Number (ISIN) — a twelve-character alphanumeric code assigned by national numbering agencies under ISO 6166 — provides a globally recognized instrument identifier. The Legal Entity Identifier (LEI), maintained by the Global Legal Entity Identifier Foundation (GLEIF) under ISO 17442, provides unique identification for legal entities including issuers, counterparties, and intermediaries. Both systems are well-established, globally adopted, and institutionally governed.

However, neither ISIN nor LEI carries semantic state. An ISIN identifies an instrument but does not carry its compliance classification, its settlement semantics, its current lifecycle state, or any record of its multi-rail representations. An LEI identifies an entity but does not encode that entity's role in any particular tokenization arrangement. Neither standard provides a resolution mechanism that returns a structured, machine-readable semantic record that a tokenization platform can use to deterministically configure itself when receiving a token. The proposed Canonical Asset Identifier (CAI) is designed to fill this gap, with ISIN and LEI serving as foundational anchoring references within the CAI record.

4.2 Architecture of a Canonical Identity Record

A Canonical Identity Record (CIR) is the authoritative semantic record for a tokenized asset. It is the single source of truth to which all rail-specific representations are bound. The schema below defines the canonical record structure:

```
// Canonical Identity Record — Schema v1.0

{
  canonicalID
  :
  "CAI-{jurisdiction}-{assetClass}-{uuid}"
  ,
  // globally unique persistent identifier
  assetClass
  :
  "debt | equity | commodity | derivative | CBDC | receivable"
  ,
  legalJurisdiction
  :
  "ISO 3166-1 alpha-2 code of primary legal domicile"
  ,
  issuingEntity
  :
  "GLEIF LEI of the issuing legal entity"
```

```
,  
  
  instrumentISIN  
  
  :  
  
  "ISO 6166 ISIN (if applicable)"  
  
  ,  
  
  representationRails  
  
  : [  
  
  {  
  
    railID  
  
    :  
  
    "string: unique identifier of the rail"  
  
    ,  
  
    railType  
  
    :  
  
    "DLT | CBDC | CSD | Payment | ISO20022"  
  
    ,  
  
    railLocalID  
  
    :  
  
    "rail-specific token or account identifier"  
  
    ,  
  
    bindingProof  
  
    :  
  
    "cryptographic binding attestation"
```

,

representationStatus

:

"active | suspended | transferred"

}

],

complianceAnchor

:

"pointer to versioned Compliance Semantic Record"

,

settlementProfile

:

"array of Settlement Instruction Ontology records per rail"

,

lifecycleState

:

"active | suspended | matured | recalled | defaulted"

,

lifecycleEvents

:

"array of timestamped lifecycle events with FMI attestation"

,

lastCanonicalUpdate

```

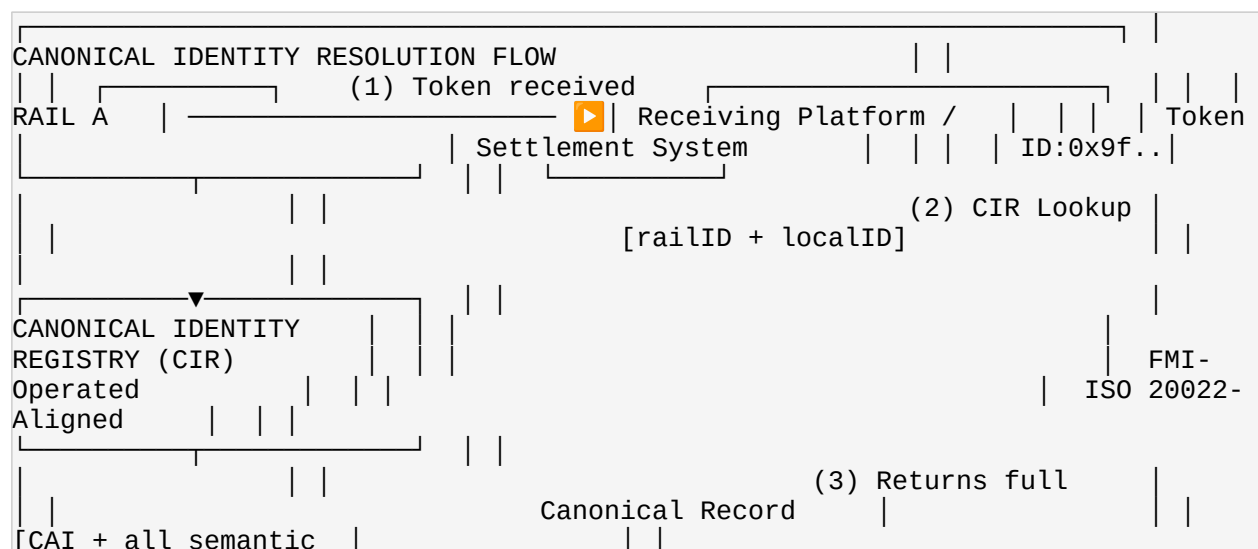
:
"ISO 8601 timestamp of last authoritative state synchronization"
,
cirVersion
:
"semantic version of this record schema"
}

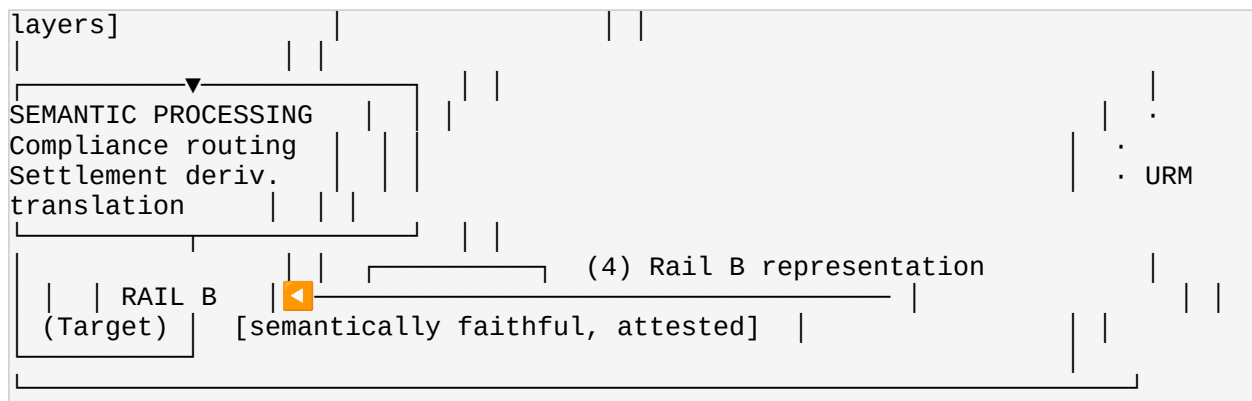
```

The `representationRails` array is the critical element that distinguishes the CIR from an ISIN record: it maintains an authoritative, cryptographically attested enumeration of every active rail representation of the asset, enabling the resolution of rail-specific identifiers to the canonical record and vice versa.

4.3 Canonical Identity Resolution Protocol

When a token is presented on any rail — whether as an on-chain object, a CSD account entry, a CBDC balance, or an ISO 20022 message payload — the receiving system initiates a canonical identity resolution to retrieve the asset's full semantic record. The resolution flow operates as follows:





The Canonical Identity Registry must be governed by an authoritative body with appropriate reach across the FMI ecosystem. The proposed governance model positions CSDs and SWIFT-connected FMIs as the natural operators of national or regional CIR nodes, federated into a global CIR network aligned with ISO 20022 data model standards and GLEIF LEI anchoring. Resolution queries use a standardized API aligned with ISO 20022 reporting message schemas, enabling legacy systems to query the CIR without requiring DLT integration.

4.4 Cross-Rail Identity Binding

Each rail-specific representation of an asset must be cryptographically or cryptographically-adjacent bound to the canonical record at the time of that representation's creation. For DLT-native rails, binding is achieved by embedding the CAI within the token's smart contract state at mint time and registering the resulting token address as an entry in the CIR's `representationRails` array, together with a hash proof of the minting transaction. For CSD account-based rails, binding is achieved through a digitally signed registration message that maps the account position identifier to the CAI. For ISO 20022 rails, the CAI is embedded as a supplementary data element in the relevant sese (securities settlement) message payload.

Update consensus requirements for the canonical record are strict: modifications to the `lifecycleState`, `complianceAnchor`, or `settlementProfile` require multi-party attestation from at least the issuing entity's CSD and the relevant regulatory reporting authority, ensuring that no single operator can unilaterally alter the authoritative semantic record of a live instrument.

5. Unified Representation Model

5.1 The Representation Problem

Financial tokenization has converged on no single data model. Four fundamentally different paradigms coexist across the FMI landscape:

- **UTXO-based models** (Bitcoin-derived systems): Assets are represented as unspent transaction outputs; ownership is implicit in transaction history; there is no persistent state object representing a position.
- **Account-based models** (Ethereum-derived, CBDC systems): Assets are represented as balances in account state; ownership is explicit in account identity; smart contract state maintains asset properties.
- **Message-based models** (ISO 20022): Assets are represented through structured instruction and confirmation messages; position records are maintained in external systems; the message is not the asset, merely an instruction regarding it.
- **Object-based models** (smart contract state, CSD proprietary schemas): Assets are represented as structured objects with explicit property fields managed by platform-specific data models.

Translation between these paradigms today is performed by bilateral integration bridges that are syntactically focused: they map fields between schemas without verifying that the translated representation preserves the full semantic meaning of the original. The Unified Representation Model (URM) is proposed as the semantic translation substrate that makes such translation meaning-preserving and formally verifiable.

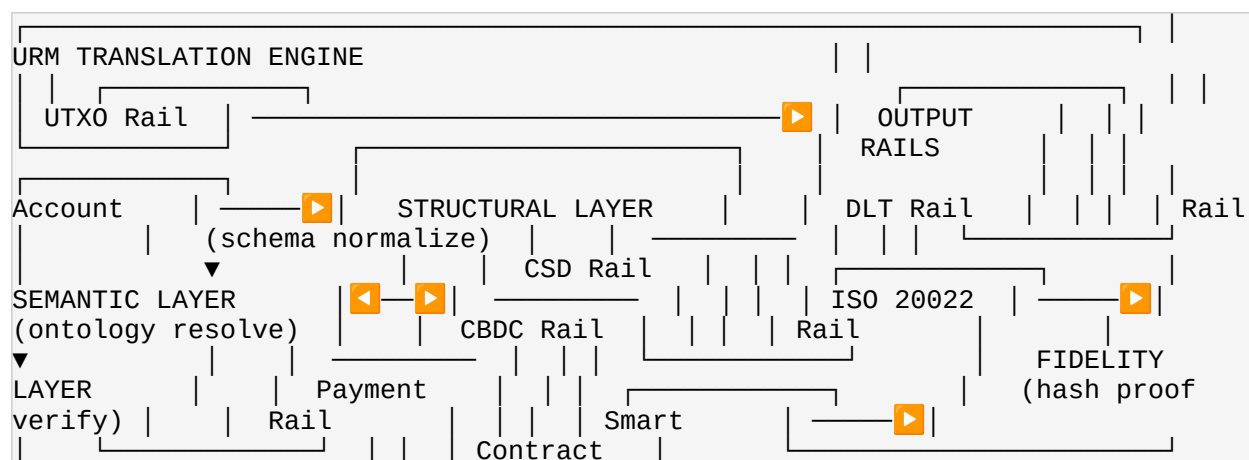
5.2 URM Architecture

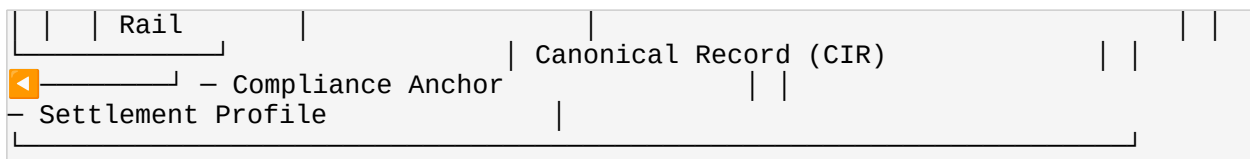
The URM operates across three sub-layers:

(a) Structural Layer: Schema normalization. The structural layer maintains a canonical JSON-LD-compatible schema for each asset class, defining the minimum and extended property sets required for a complete semantic representation. When a rail-specific representation is received, the structural layer maps it into this canonical schema, identifying any fields present in the source that have no canonical equivalent (potential semantic loss) and any canonical fields absent from the source (potential semantic deficit).

(b) Semantic Layer: Meaning-preserving translation. The semantic layer resolves ambiguities that purely structural mapping cannot: cases where two different source fields encode the same semantic concept, cases where a single source field encodes multiple semantic concepts, and cases where the semantic meaning of a field depends on context that is not encoded in the field itself. The semantic layer uses an asset-class ontology — a formal specification of the semantic relationships between financial instrument properties — to perform these resolutions correctly.

(c) Fidelity Layer: Verification of semantic completeness. The fidelity layer computes a hash of the canonical representation's semantic core — the set of properties that determine the economic claims, compliance obligations, and settlement instructions attached to the asset — and verifies that this hash is preserved across translation. A translation is rejected if it alters any element of the semantic core without an explicit, attested justification.





5.3 Representation Fidelity Guarantees

Formal Definition

"A representation is semantically faithful if and only if the canonical identity of the represented asset resolves to identical economic claims, compliance states, and settlement obligations regardless of the rail from which the resolution is initiated, and the fidelity hash of the semantic core is invariant across all active representations."

Fidelity proofs are implemented as a hash chain anchored to the CIR: when a new representation is created on a new rail, the URM computes the fidelity hash of its semantic core and registers it as an entry in the canonical record. Any subsequent representation claiming the same canonical identity must present a fidelity hash that matches this registered value to within the tolerance permitted by the canonical schema (allowing for rail-specific representation differences in non-core fields, such as format or encoding). Oracle attestation patterns — in which trusted third parties (typically the issuing CSD or a designated verification authority) co-sign fidelity claims — provide an additional verification layer for high-value instruments.

6. Multi-Rail Deterministic Settlement

6.1 The Settlement Semantics Problem

Settlement — the final transfer of an asset from seller to buyer against the simultaneous transfer of cash consideration — is the operational heart of financial market infrastructure. The systemic importance of settlement finality is reflected in the CPMI-IOSCO Principles for Financial Market Infrastructures (PFMIs), which establish unambiguous finality and timely settlement as core requirements for systemically important FMIs. Yet the current tokenization landscape has systematically reproduced the settlement ambiguities of legacy systems rather than resolving them.

In current tokenized systems, settlement routing is determined procedurally at the application layer: an operator configures which settlement rail to invoke for a given transaction type, asset class, or counterparty pair. This configuration is platform-specific, not semantically derived from the asset itself. The result is settlement non-determinism: the same DVP instruction, for the same instrument, against the same counterparty, may settle differently depending on which platform initiates it. Across rail boundaries, where delivery and payment legs may be hosted on different systems, atomic settlement becomes structurally impossible without an explicit coordination protocol — which current syntactic bridges do not provide.

6.2 Multi-Rail Settlement Semantic Model

Definition

"Settlement determinism": given a canonical asset and a valid settlement instruction, the correct settlement rail, settlement protocol, DVP mechanism, and finality standard are uniquely determined by the asset's embedded semantic settlement properties — not by operator configuration. The settlement outcome is identical regardless of which compliant platform initiates the instruction.

Settlement semantics are embedded in the token's canonical record as a structured Settlement Profile, which is an array of Settlement Instruction Ontology records — one per active rail representation. Each record encodes the settlement parameters applicable to that rail, derived from the asset's legal, regulatory, and operational characteristics at the time of canonical record creation, and updatable only with authorized attestation.

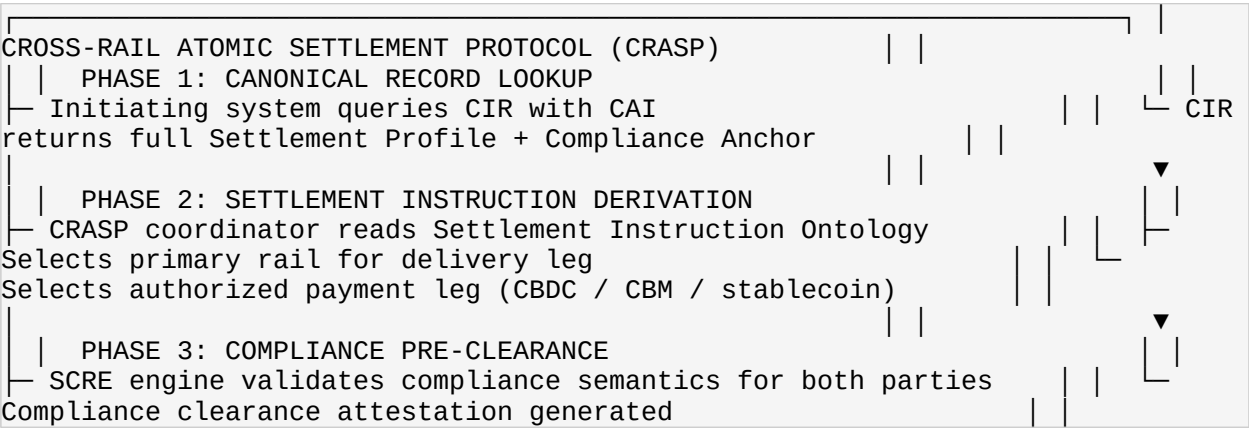
6.3 The Settlement Instruction Ontology

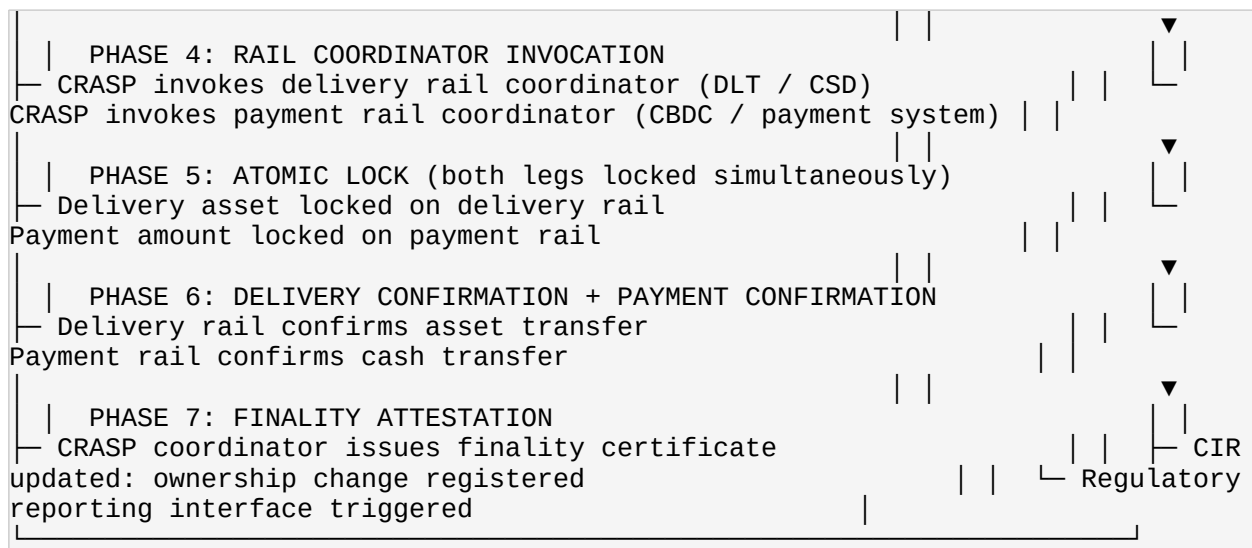
Ontology Field	Type	Description	Example Values
<code>settlementRailPrimary</code>	string	Preferred settlement rail identifier for this asset class and jurisdiction	"T2S-ECB", "DTCC-DTC", "HKMA-CBDC"
<code>settlementRailFallback</code>	array[string]	Ordered list of fallback rails if primary is unavailable; defines fallback precedence	["CSD-PROP", "CORR-BANK"]
<code>settlementFinality</code>	enum	Finality standard applicable to this asset: gross (each transaction final individually), net (final at end of netting cycle), or conditional (subject to DvDvP conditions)	gross / net / conditional
<code>paymentLeg</code>	enum	Authorized payment leg instruments for DVP against this asset	CBDC / commercial-bank-money / stablecoin / correspondent
<code>deliveryLeg</code>	enum	Delivery mechanism for the asset leg of the DVP transaction	DLT-native / CSD-native / hybrid

Ontology Field	Type	Description	Example Values
atomicityRequirement	boolean	Whether the delivery and payment legs must settle atomically (both or neither)	true / false
nettingEligible	boolean	Whether positions in this asset are eligible for multilateral netting	true / false
finalityAnchor	enum	Standard settlement cycle defining when finality is achieved	T+0 / T+1 / T+2 / real-time-gross
dvcRequirements	object	Delivery versus payment coordination requirements including pre-matching, confirmation, and lock-up protocols	{prematching: required, confirmWindow: 4h}

6.4 Cross-Rail Atomic Settlement Protocol (CRASP)

The Cross-Rail Atomic Settlement Protocol (CRASP) is proposed as the coordination layer that achieves atomic finality across heterogeneous rails using semantically embedded settlement instructions. CRASP operates as a seven-phase protocol:





CRASP resolves the atomic settlement problem through the lock mechanism in Phase 5: both legs are placed in a coordinated locked state before either is irrevocably committed. If either lock fails, both are released without settlement occurring, preserving the no-settlement-without-both-legs guarantee. This is the cross-rail equivalent of the Delivery versus Payment principle enshrined in the CPMI PFMI, extended to function across heterogeneous settlement systems.

7. Semantic Compliance Routing

7.1 Compliance as Intrinsic Semantics

The compliance architecture of current tokenization platforms reflects the procedural compliance workflows of legacy financial systems: compliance checks are performed as gates at transaction initiation, by external services that interrogate the transaction, the token, and the counterparties, and return an authorization decision. This architecture was inherited from pre-tokenization systems where compliance logic was necessarily separated from asset representation, because assets were records in closed, system-specific databases rather than portable digital objects.

Semantic Tokenization inverts this architecture. Compliance obligations are not extrinsic gates applied to a token — they are intrinsic semantic properties of the token itself. The token "knows," as a structural attribute of its own definition, what compliance checks govern each class of transaction, what jurisdictions' regulatory regimes apply, what counterparty characteristics are required, and how compliance state must be reported. This approach enables compliance routing to be automated, consistent, and machine-readable by regulators — eliminating the compliance gap that currently arises when assets cross platform or rail boundaries.

7.2 Compliance Semantic Schema

Schema Field	Type	Description
complianceJurisdictions	array[string]	Array of applicable regulatory regimes by identifier (e.g., "EU-MiCA", "US-SEC-SA33", "MAS-SFA", "EU-DORA", "UK-FCA-FSMA")
kycRequirement	enum	Minimum KYC tier required for ownership: standard (natural persons / retail), enhanced (high-risk counterparties), institutional-only (QIB / professional investor only)
amlRiskClass	enum	AML risk classification of the asset and instrument type: low / medium / high / prohibited
sanctionsScreeningRequired	boolean	Whether OFAC / EU / UN sanctions screening is required at each transfer for both delivery and payment legs
suitabilityConstraints	enum	Investor eligibility restrictions: retail-eligible / professional-only (MiFID II) / institutional-only (QIB) /

Schema Field	Type	Description
		accredited-only
transferRestrictions	enum + config	Transfer restriction model: unrestricted, whitelist (only pre-approved counterparties), jurisdictional-fence (transfers restricted to specified jurisdictions)
complianceAnchorVersion	string	Versioned pointer to the compliance ruleset document, enabling compliance logic versioning without asset re-issuance
reportingObligations	array[object]	Array of regulatory reporting obligations triggered by each transaction type, including report schema, recipient authority, and timing requirement

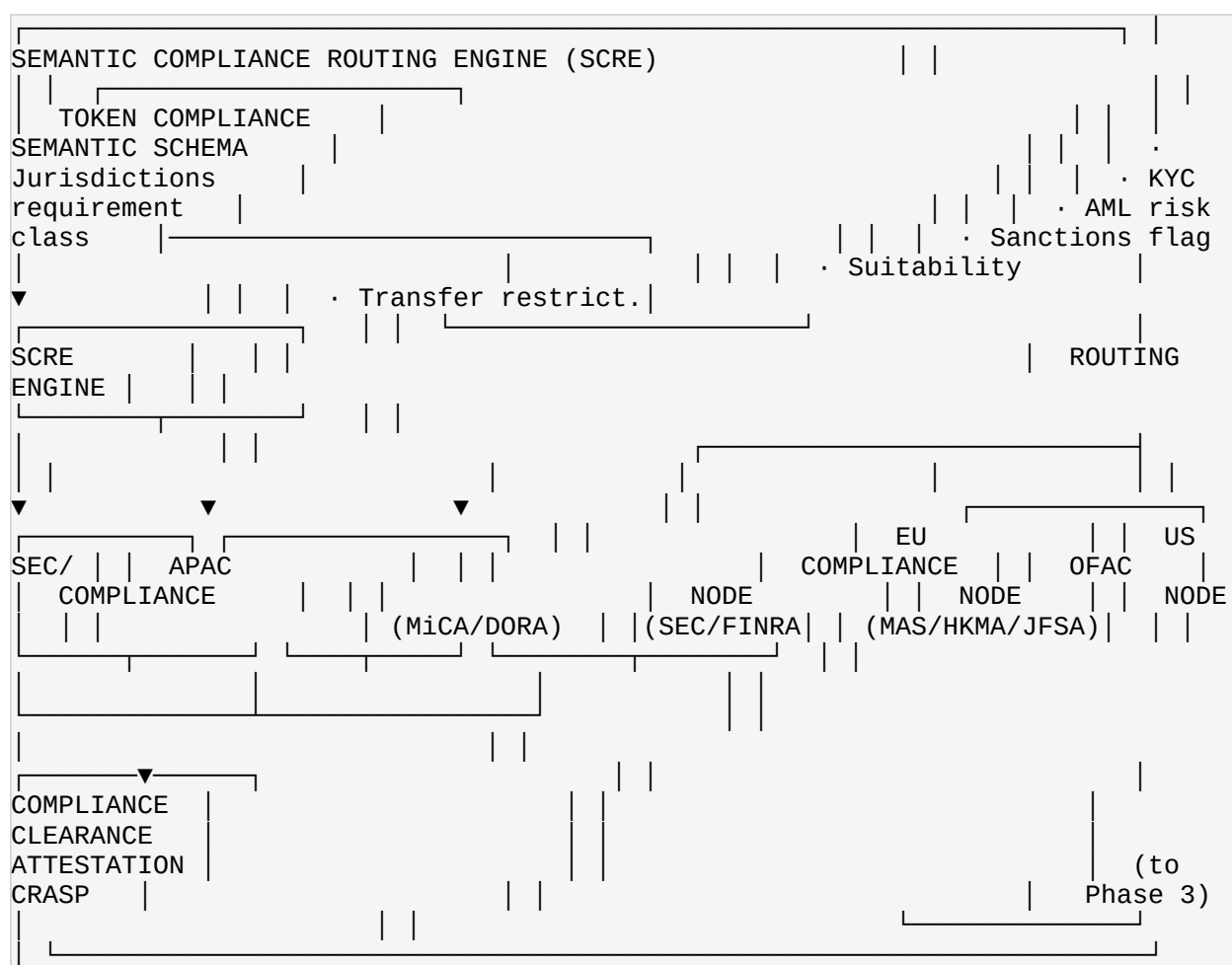
7.3 Semantic Compliance Routing Engine (SCRE)

The Semantic Compliance Routing Engine (SCRE) is the operational layer that reads the token's compliance semantic schema and automatically routes the compliance verification workflow to the correct jurisdiction-specific compliance infrastructure at transaction initiation. SCRE eliminates three categories of compliance failure:

- Duplicate KYC across rails:** SCRE maintains a cross-rail KYC clearance registry indexed to the canonical compliance anchor. A counterparty cleared under the applicable KYC tier on one rail need not be re-cleared on other rails where the same compliance anchor applies, subject to currency controls (clearances expire according to the applicable jurisdiction's refresh requirements).
- Jurisdictional compliance gaps:** When an asset crosses rail or jurisdictional boundaries, SCRE reads the `complianceJurisdictions` array and

identifies any new jurisdictions implicated by the cross-rail transfer that were not implicated by the originating transaction. It initiates compliance routing for these new jurisdictions proactively, before the cross-rail handshake completes.

- **Manual compliance configuration:** Because compliance routing is derived from the token's semantic schema, platform operators do not configure compliance logic for each instrument. The asset carries its own compliance instructions, and SCRE executes them uniformly across all platforms that implement the SCRE protocol.



7.4 Regulatory Machine Readability

A consequence of embedding compliance semantics intrinsically in the token is that regulatory supervisory access is fundamentally transformed. Today, supervisors reconstruct compliance status by examining transaction logs, querying custodians,

and reviewing disclosure submissions — a laborious, time-lagged, and necessarily incomplete process. When compliance semantics are embedded in the canonical record, supervisors can query the CIR directly to obtain a machine-readable, structured representation of the compliance status of any instrument at any point in time.

This aligns with the direction of travel in EU Digital Finance regulation: the Markets in Crypto-Assets (MiCA) regulation, in force since December 2024, establishes requirements for structured disclosure and regulatory reporting for crypto-asset service providers and issuers. The EU Digital Operational Resilience Act (DORA), applicable from January 2025, imposes operational resilience reporting requirements on digital financial infrastructure. Semantic compliance properties embedded in the token provide the data substrate required to fulfill these obligations programmatically, reducing the compliance cost burden on FMIs while improving the accuracy and timeliness of supervisory data.

8. Cross-Rail Semantic Interoperability

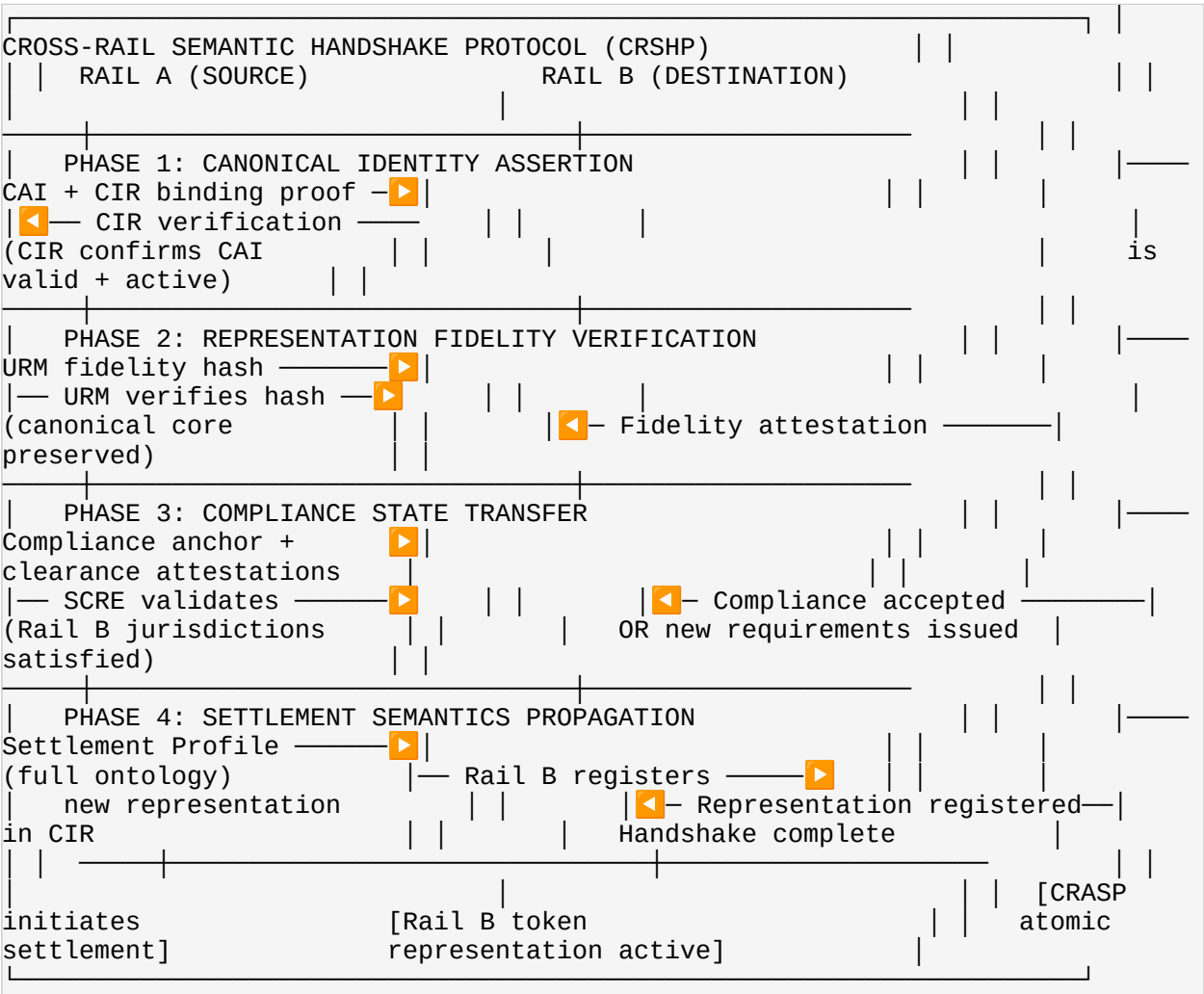
8.1 The Interoperability Gap

The financial infrastructure community has invested significantly in syntactic interoperability solutions: atomic swap protocols, cross-chain messaging systems (such as the Inter-Blockchain Communication protocol), and SWIFT-connected DLT bridges. These solutions move tokens or token representations across system boundaries, but they do so syntactically: they serialize the token's data in the source format, transmit it, and deserialize it in the destination format. They do not verify that the deserialized representation is semantically equivalent to the source — that it represents the same economic claim, carries the same compliance state, and will settle the same way.

The FSB's 2024 report on tokenization explicitly identifies lack of interoperability among DLT platforms and between DLT platforms and traditional financial infrastructure as a principal barrier to meaningful tokenization scaling. Semantic interoperability addresses this gap by requiring that cross-rail transfers preserve not only the token's data structure but its full semantic content — canonical identity, compliance state, representation fidelity, and settlement semantics — with cryptographic attestation that no semantic loss has occurred in transit.

8.2 Cross-Rail Semantic Handshake Protocol

The Cross-Rail Semantic Handshake Protocol (CRSHP) defines the four-phase semantic coordination sequence that must be completed before any cross-rail asset transfer achieves legal and operational finality:



Only upon successful completion of all four phases does the cross-rail transfer proceed to CRASP for atomic settlement. If any phase fails — if canonical identity cannot be verified, if fidelity is compromised, if compliance requirements for the destination rail cannot be satisfied, or if settlement semantics cannot be propagated — the handshake aborts and no asset transfer occurs, preserving systemic integrity.

8.3 Semantic Bridge Architecture

A semantic bridge differs from a syntactic bridge in the same way that a professional legal translator differs from a word-for-word dictionary translator: the former preserves meaning; the latter preserves form. A semantic bridge comprises four operational components, each corresponding to one phase of the CRSHP handshake:

- **Canonical ID Resolver:** Verifies that the source rail's local token identifier resolves to a valid, active Canonical Asset Identifier via the CIR. Manages resolution caching and freshness requirements.
- **URM Translation Engine:** Performs the three-layer URM translation (Structural → Semantic → Fidelity) to produce a destination-rail-compatible representation without semantic loss.
- **Compliance State Verifier:** Reads the compliance anchor from the canonical record, verifies that existing compliance clearances satisfy the destination rail's jurisdictional requirements, and identifies any additional compliance verifications required.
- **Settlement Instruction Propagator:** Reads the Settlement Profile from the canonical record and registers a new Settlement Instruction Ontology entry for the destination rail, incorporating any rail-specific settlement parameter adjustments within the bounds permitted by the asset's settlement semantics.

8.4 FMI Interoperability Implications

The relevance of semantic interoperability for existing FMI initiatives is concrete and immediate. BIS Project mBridge, which reached minimum viable product status in mid-2024 before being handed over to its participant central banks, demonstrated the technical feasibility of peer-to-peer cross-border CBDC settlement across multiple central bank platforms. However, mBridge operates at the payment rail level — it moves CBDC value but does not carry the semantic properties of the assets being settled. Integration with a Semantic Tokenization layer would enable mBridge-style infrastructure to serve as the payment leg of semantically deterministic DVP transactions, with CBDC settlement invoked by the CRASP protocol rather than by operator configuration.

The ECB's DLT trials within the T2S framework, conducted through 2024 and 2025, explored the integration of DLT-based settlement with the established CSD ecosystem. These trials exposed the critical need for a canonical identity layer: DLT-issued securities and T2S-settled securities of the same underlying instrument required manual mapping rather than automated semantic resolution. A CIR-based canonical identity layer would have resolved this automatically.

FedNow, the Federal Reserve's real-time gross settlement system launched in 2023, provides an always-available payment rail with immediate finality characteristics that align naturally with the T+0 and real-time-gross finality anchors in the Settlement Instruction Ontology. Semantic integration between FedNow and DLT-based delivery rails would enable real-time atomic DVP for US-dollar-denominated tokenized securities — a capability currently achievable only through manual coordination between settlement systems.

9. Implications for Financial Market Infrastructure

9.1 For Central Securities Depositories (CSDs)

CSDs are the natural institutional operators of Canonical Identity Registries. Their existing mandate — as the authoritative registrars of securities ownership and the operators of settlement infrastructure — gives them both the institutional legitimacy and the operational capability to maintain canonical records for the instruments they serve. Euroclear, Clearstream, the DTC, and JASDEC already maintain the most authoritative records of instrument existence and ownership in their respective jurisdictions; the CIR is, conceptually, an enriched and semantically structured evolution of the records they already maintain.

However, the CSD's role in a semantic tokenization ecosystem requires a fundamental conceptual evolution: from syntactic record-keeper to *semantic asset anchor*. A syntactic record-keeper records who holds what; a semantic asset anchor maintains the authoritative meaning of the asset — its legal character, its compliance obligations, its settlement semantics — in a form that is machine-readable, computationally actionable, and trustworthy across the full multi-rail ecosystem. CSDs that make this evolution position themselves as indispensable infrastructure for the tokenized economy; those that do not risk disintermediation by DLT-native platforms that create their own, lower-governance canonical records.

9.2 For Central Counterparties (CCPs)

Semantic settlement instructions enable CCPs to perform semantically-aware netting: rather than configuring netting eligibility instrument-by-instrument through manual rulebooks, CCPs can read the `nettingEligible` and `settlementFinality` properties from the Settlement Instruction Ontology embedded in each token's canonical record. Netting eligibility becomes a property of the asset, verifiable in real time, rather than a configuration maintained in CCP systems that must be manually updated when instrument characteristics change.

Cross-rail margin and collateral management presents a further opportunity. When collateral posted on one rail is represented in the CIR with its full semantic profile including asset class, jurisdiction, and credit quality, a CCP's margin system can evaluate the collateral's adequacy without requiring platform-specific integration

with each rail's data model. The URM translation layer ensures that collateral valuation and haircut calculation operates on semantically consistent representations regardless of the source rail.

9.3 For CBDC Platforms

Wholesale CBDC platforms — including those developed under BIS Project mBridge, the ECB's wholesale CBDC trials, and the Federal Reserve's Project Hamilton explorations — are natural payment leg candidates for the CRASP settlement protocol. When a CBDC platform is registered in a token's Settlement Instruction Ontology as an authorized payment leg, CRASP can invoke it deterministically for settlement without requiring operator-level configuration of the payment rail selection.

The interoperability between wholesale CBDC (wCBDC) and commercial bank money within a unified semantic settlement model is a particularly significant challenge. In a wCBDC-enabled DVP transaction, the payment leg settles in central bank money — achieving the highest achievable finality — while the delivery leg may settle on a DLT rail or through a CSD. Semantic tokenization provides the coordination substrate: the CRASP protocol treats wCBDC settlement as a first-class payment leg option specified in the Settlement Instruction Ontology, enabling the payment system to be invoked atomically with the delivery system.

9.4 For Tokenization Platforms

Private tokenization platforms — those issuing tokenized equities, bonds, structured products, fund units, and real-world assets — must adopt canonical identity binding at token genesis rather than treating identity as a post-issuance concern.

Retrofitting canonical identity to an existing population of tokens is structurally difficult: it requires matching existing local identifiers to canonical records without an authoritative binding proof at mint time, relying instead on the platform operator's attestation. Such attestations carry weaker trust guarantees than binding proofs generated and registered at mint time.

The operational implication is clear: tokenization platforms should integrate CIR registration as the first step of their token minting workflow. At the moment a tokenized instrument is created, the platform should simultaneously register the canonical record in the CIR, bind the local token identifier to the CAI, and populate the Settlement Profile and Compliance Semantic Schema from the instrument's legal documentation. This "semantic-first" minting approach produces tokens that are immediately interoperable across the semantic tokenization ecosystem rather than requiring subsequent integration effort.

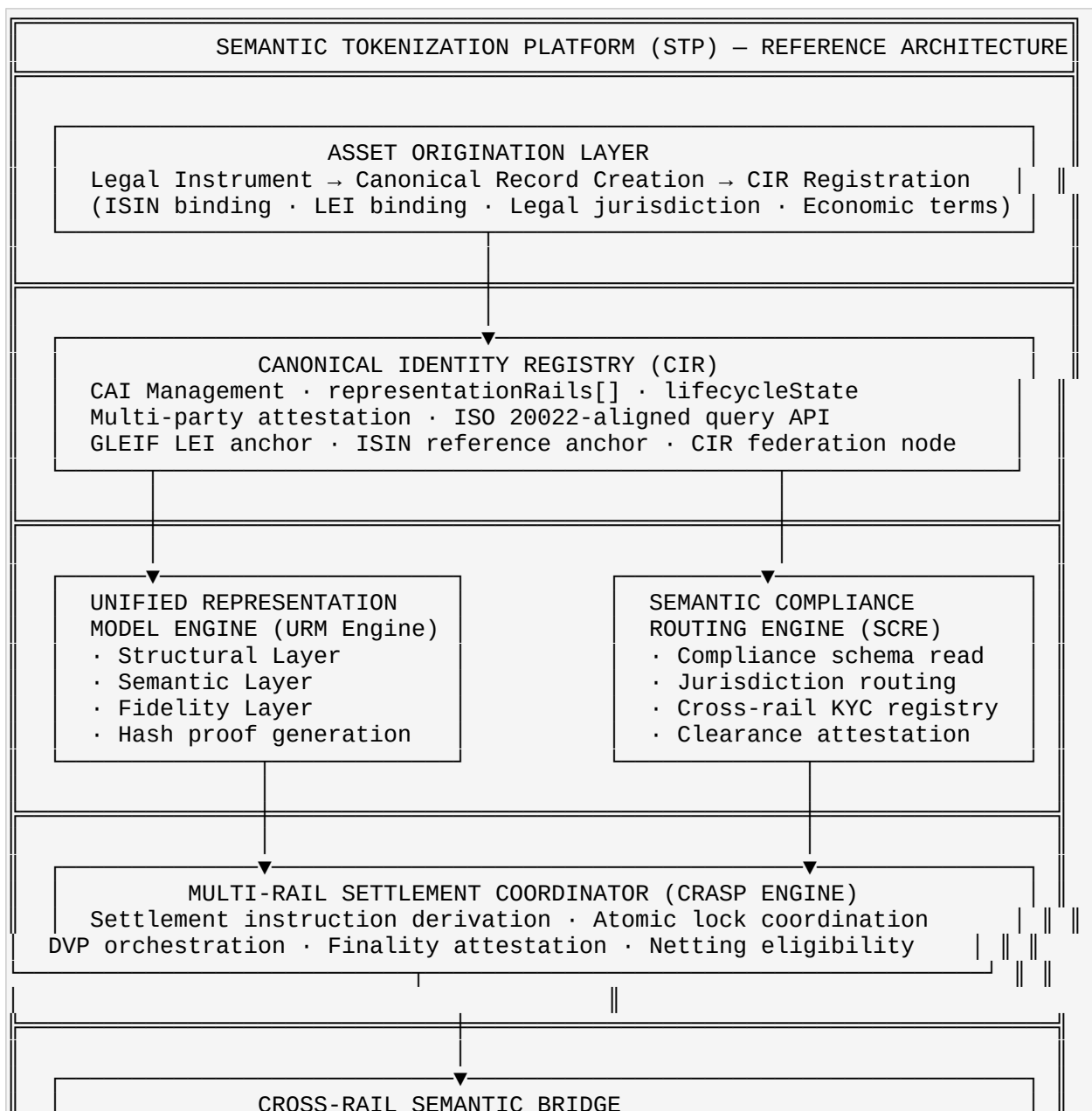
9.5 For Regulators and Policy Bodies

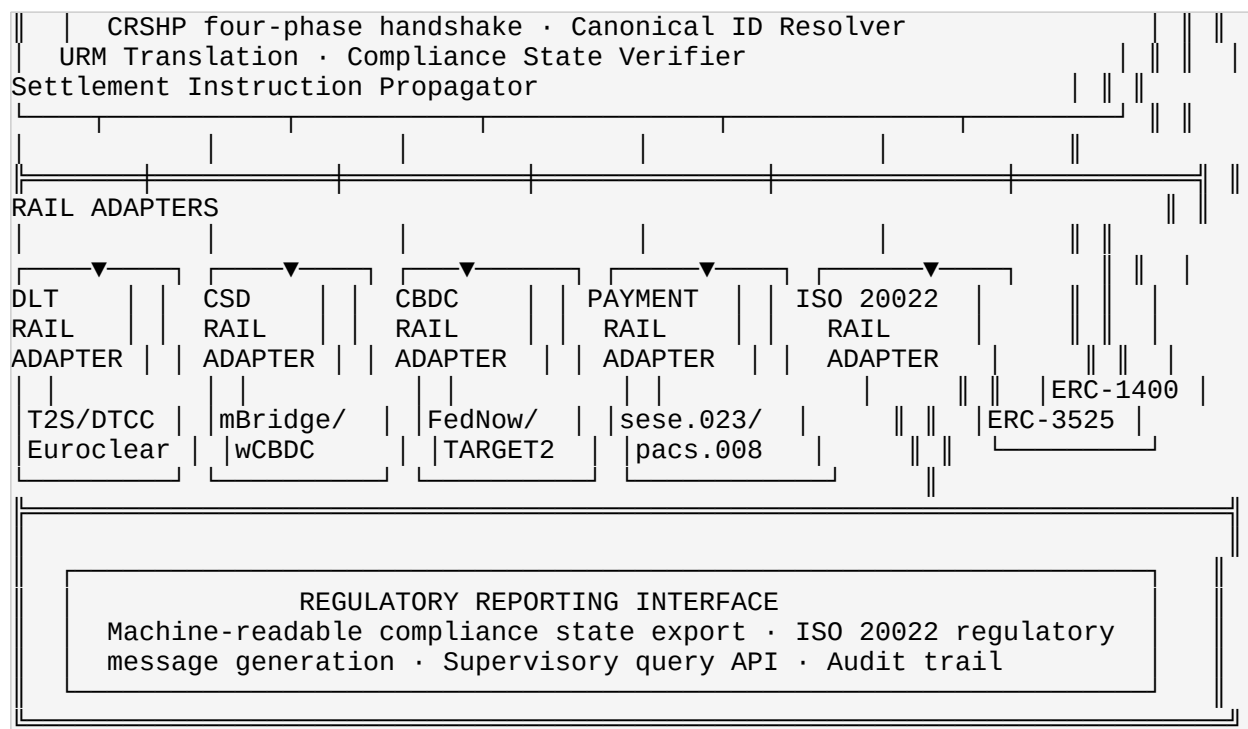
Machine-readable compliance semantics embedded in canonical token records provide regulators with a fundamentally improved supervisory tool. Rather than reconstructing an instrument's compliance status from fragmented transaction logs and custodian disclosures, a regulator with access to the CIR can query the authoritative compliance anchor for any instrument and receive a structured, versioned, and attested compliance record in real time. This capability aligns with the regulatory objectives articulated in EU MiCA, DORA, and the FSB's tokenization policy recommendations.

Semantic tokenization also provides the foundation for programmable regulatory compliance — the concept that compliance obligations can be encoded as executable logic that is automatically applied to transactions, eliminating the need for manual compliance review of routine transactions while preserving human judgment for complex or anomalous cases. This does not require that compliance decisions be made by algorithms; rather, it requires that the inputs to compliance decisions — the asset's risk classification, the counterparties' KYC status, the applicable jurisdictional requirements — be structured, machine-readable, and consistently applied.

10. Reference Architecture: Semantic Tokenization Platform (STP)

The following reference architecture integrates all five semantic layers into a complete Semantic Tokenization Platform (STP). The architecture is organized vertically from asset origination at the top to rail adapters at the bottom, with horizontal integration layers spanning all rail types.





The data flow through this architecture follows the sequence of semantic layers defined in Section 3. An asset is originated at the Asset Origination Layer, where its legal instrument properties are read and a Canonical Identity Record is created and registered in the CIR. The URM Engine and SCRE are populated with the asset's representation schema and compliance schema respectively. When a transaction is initiated involving the asset, the CRASP Engine reads the Settlement Instruction Ontology from the CIR, coordinates with the SCRE for compliance pre-clearance, invokes the Cross-Rail Semantic Bridge if rail boundaries are crossed, and executes atomic settlement through the appropriate Rail Adapters. All settlement events, lifecycle changes, and compliance attestations are propagated to the Regulatory Reporting Interface for supervisory access.

11. Governance, Standards, and Adoption Pathways

11.1 Standards Alignment

The Semantic Tokenization framework is designed to be complementary to, and ultimately implemented within, existing financial messaging and identification standards. ISO 20022 — the international standard for electronic data interchange between financial institutions, mandated for SWIFT cross-border payments from November 2025 — provides the richest available semantic messaging substrate for financial transactions. Its data model, which specifies structured, typed, and validated fields for parties, instruments, amounts, and instructions, is the natural encoding format for the Unified Representation Model's canonical schema. ISO 20022 already supports securities settlement messaging (sese domain), securities clearing (setr domain), and payments (pacs domain) — the three primary transaction types that semantic tokenization must coordinate.

GLEIF's LEI system, operating under ISO 17442, provides the entity identity anchor for the Canonical Identity Record's `issuingEntity` field and for counterparty identification in SCRE compliance routing. The LEI's verified data — including the legal entity's registered name, jurisdiction, and ownership structure — can be directly incorporated into the compliance semantic schema to support KYC and suitability determination.

The critical standards gap that this paper identifies is the absence of a Canonical Asset Identifier (CAI) standard. Neither ISIN nor any existing identifier provides the semantic record resolution capability required by the CIR. We propose that a CAI standard be developed through the ISO TC68 (Financial Services Technical Committee) working group structure, in collaboration with GLEIF, ANNA (Association of National Numbering Agencies, the ISIN governance body), and the BIS's Committee on Payments and Market Infrastructures (CPMI). The CAI should be specified as an extension of the existing LEI and ISIN namespaces rather than an entirely new identification system, minimizing adoption friction for institutions already using these standards.

Standards Gap Note

A critical observation: ISO 20022's rich data model provides semantic messaging capability but does not define persistent semantic asset identity. ISIN identifies instruments but carries no semantic state. LEI identifies entities but not their tokenization roles. The CAI standard fills the gap at the intersection of all three, providing a persistent, semantically rich, resolvable asset identifier for the tokenized financial system.

11.2 Governance Models

Three governance architectures for the Canonical Identity Registry are feasible, each with distinct trust, resilience, and operational characteristics:

Governance Model	Operator	Trust Model	Resilience	Suitability
Centralized CIR	Single FMI operator (e.g., SWIFT, global CSD)	High institutional trust; single authority	Single point of failure; high concentration risk	Appropriate for early pilots; not suitable for systemic infrastructure
Federated CIR	Consortium of national/regional FMIs (CSDs, central banks)	Distributed institutional trust; consensus-based updates	Resilient; no single point of failure; requires consensus governance	Recommended for wholesale financial infrastructure
Decentralized CIR	DLT-native consensus network	Algorithmic trust; no single institutional authority	Highly resilient; complex governance; censorship-resistant	May be appropriate for global retail-accessible instruments; high governance complexity

The federated model is recommended for the wholesale financial infrastructure context addressed by this paper. A federation of established FMIs — CSDs, central

banks, SWIFT — provides the institutional legitimacy required for systemic-scale adoption, the governance clarity required for regulatory recognition, and the distributed architecture required to prevent concentration risk. The governance framework for a federated CIR should be modeled on the established governance of existing multi-party FMIs: rule book-based, with defined participant obligations, update procedures, and dispute resolution mechanisms. The T2S governance framework, BIS mBridge's bespoke decentralized governance model, and GLEIF's federated validation agent network provide useful reference points.

11.3 Adoption Pathway

A realistic adoption pathway for Semantic Tokenization at global FMI scale requires a phased approach, recognizing both the technical complexity of the framework and the institutional dynamics of FMI ecosystem change. The following four-phase roadmap is proposed:

- **Phase 1 (2026–2027): Canonical Identity Layer Adoption.** Major CSDs and tokenization platforms adopt the CAI standard and begin registering canonical records for newly issued tokenized instruments. CIR pilot nodes are established by two or three leading CSDs in coordination with their national central banks. ISO TC68 CAI working group is constituted and produces initial draft standard.
- **Phase 2 (2027–2028): Compliance Semantics Standardization.** Compliance Semantic Schema is standardized through ISO TC68 and IOSCO technical working groups. SCORE reference implementations are developed by major compliance technology vendors. EU MiCA compliance reporting integration with SCORE is piloted by European tokenization platforms.
- **Phase 3 (2028–2030): Multi-Rail Settlement Semantic Integration.** CRASP protocol is implemented by leading settlement infrastructure operators. Wholesale CBDC platforms — including next-generation mBridge successors and FedNow — register as CRASP-compatible payment leg rails. Cross-border DVP pilots using semantically embedded settlement instructions are conducted.

- **Phase 4 (2030 and beyond): Full Cross-Rail Semantic**

Interoperability. Global FMI ecosystem achieves semantic interoperability at systemic scale. Regulatory reporting interfaces enabled by machine-readable compliance semantics provide real-time supervisory access. Tokenized asset markets operate with full canonical identity resolution, semantically deterministic settlement, and automated compliance routing across all major FMI rails.

12. Conclusion

This paper has argued that the current generation of financial tokenization, despite its considerable technical sophistication, is semantically impoverished. Tokens as currently implemented are syntactic artifacts: they encode transfer logic and balance data, but they do not encode the legal meaning, compliance obligations, settlement semantics, or cross-rail identity of the assets they purport to represent. The consequence is a financial infrastructure landscape in which the same government bond, equity share, or trade receivable exists simultaneously as multiple unrelated objects across incompatible systems — with no authoritative mechanism to assert their equivalence, propagate their lifecycle events, or coordinate their settlement.

Semantic Tokenization is the architectural response to this deficit. By encoding canonical identity, compliance semantics, settlement instructions, and representation fidelity guarantees as intrinsic token properties rather than extrinsic operator configurations, Semantic Tokenization transforms the token from a ledger entry into a meaning-bearing, self-describing, computationally interpretable unit of financial representation. The five-layer architecture presented in this paper — from the asset reality of the underlying legal instrument through canonical identity, unified representation, multi-rail settlement, and compliance routing to cross-rail

semantic interoperability — provides a complete and internally consistent framework for this transformation.

The implications for financial market infrastructure are profound and urgent. Central securities depositories face both an opportunity and an obligation: to evolve from syntactic record-keepers into semantic asset anchors, anchoring the tokenized economy's identity layer in the institutional trust and governance legitimacy that only established FMIs can provide. Central counterparties gain the ability to perform semantically-aware netting and cross-rail collateral management, reducing gross settlement obligations and operational risk. CBDC platforms become deterministically invokable settlement legs rather than optional configuration choices. Tokenization platforms gain instant cross-rail interoperability from the moment of token genesis rather than through costly, fragile point-to-point integrations. Regulators gain machine-readable, real-time supervisory access to the compliance state of the instruments they oversee.

The central imperative of this paper is addressed equally to FMIs, tokenization platforms, and policy bodies: the establishment of canonical identity standards, unified representation models, and semantic compliance frameworks must precede — not follow — the scaling of tokenized asset markets to systemic size. The experience of the internet's early years, in which the absence of identity and security standards created vulnerabilities that persist to this day, provides a cautionary precedent. The financial system cannot afford to tokenize first and semanticize later. The foundation of meaning must be laid before the structure of scale is erected upon it.

Call to Action

FMIs, central banks, and tokenization platforms are invited to engage with the standards bodies — ISO TC68, CPMI, IOSCO — to advance the Canonical Asset Identifier specification, the Compliance Semantic Schema standard, and the Cross-Rail Atomic Settlement Protocol as international standards. The coordination window is open. The systemic opportunity is real. Semantic coherence is the

prerequisite for tokenized finance at scale.

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